

ABC Cruise Lines



Stakeholder Team

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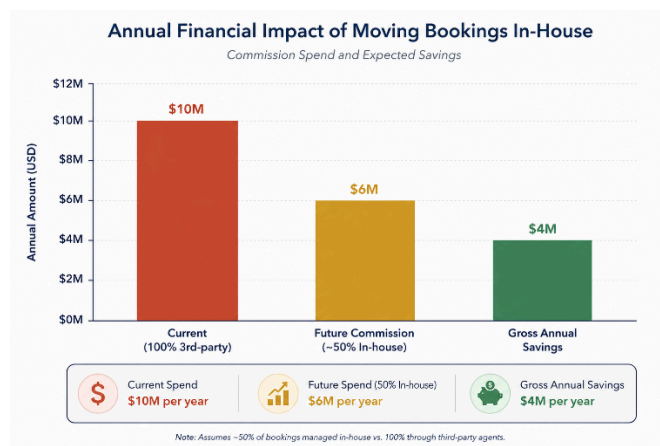
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1. Introduction & Project Objective

ABC Cruise Lines operates three ships out of New York, serving two product lines: multi-night Canadian cruises (7-night and 9-night) and 1-day / half-day “cruises to somewhere” in international waters for gambling. Currently, all reservations are booked through third-party agents at 10–15% commission, about \$10M per year, and ABC owns no reservation system or data.

This report describes the system we designed: how customers book, how booking data is used to generate a demand forecast for staffing, and how the DSS helps the Reservation Manager make two recurring operational decisions: capacity (in-house vs. third-party overflow) and pricing (raise prices or discount). A two-week case study walks the prototype end to end with sample data.

Matching the industry benchmark of ~50% self-managed bookings would cut commissions and save ~\$4M in gross annually. Beyond cost, our reservation website would allow ABC to gain ownership of customer data, provide faster service, and turn bookings into staffing and channel-mix decisions.



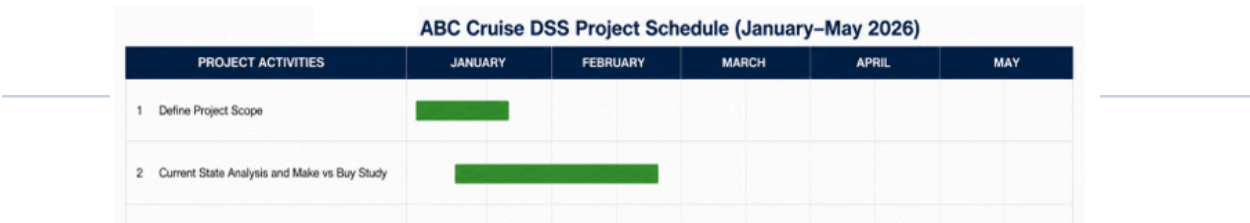
Owning the channel and the data is a durable competitive advantage: it lowers cost per booking, improves customer experience, and feeds the revenue-management and shipboard systems that the rest of the enterprise depends on.

2. Project Lifecycle

- **Project Scope:** to design and deliver a working in-house reservation website and staffing model prototype as a Decision Support System (DSS), replacing ABC's fully outsourced booking process.
- **Project Objective:** to enable ABC to capture roughly 50% of bookings directly and give the Reservation Manager data-driven control over staffing and pricing decisions.
- **Project Deliverables:** the reservation prototype, a demand-forecasting and staffing model, a manager decision dashboard, a sample demand dataset.
- **Project Budget:** is a one-time investment of approximately \$2 million, justified by roughly \$4 million in projected gross annual commission savings.
- **Project Schedule:** five months (January–May), structured around initiation and system design, prototype build, testing, and go-live.
- **Principal Stakeholders:** the Director of Operations (sponsor), the Reservation Manager (primary user), analysts, and reservation agents, with the project team.
- **Project Risks:** forecast inaccuracy, unrealistic handle-time assumptions, and slow early adoption are managed through monthly recalibration, adjustable model parameters, and phased rollout.

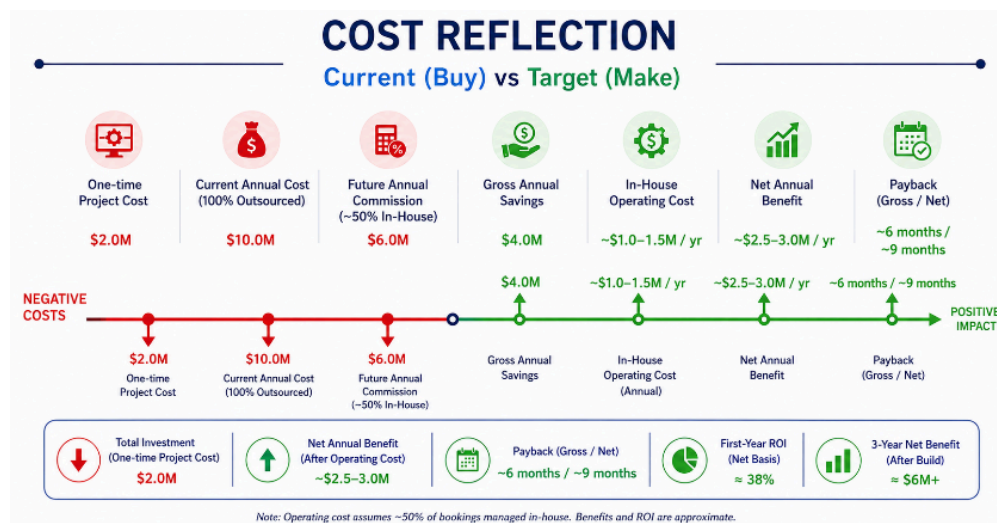
3. Project Timeline

The project runs January through May, ending at go-live in the middle of May 2022.



4. Business Case Justification

Bringing reservations in-house is a make-vs-buy decision. Starting from the financial view, even if ABC captured only 30% in-house instead of 50%, savings would still be ~\$2.4M gross—well above the ~\$2M build cost in year one. The case is robust to conservative assumptions. This business case rests on shifting reservation volume from a commissioned channel to an owned one. The figure below summarizes the annual economics.



5. Decision Support System Design

The system is a three-tier application. Customers self-serve on a booking website; an analyst cleans the booking data and then feeds it to the staffing model for staffing prediction; the manager makes decisions based on the results.

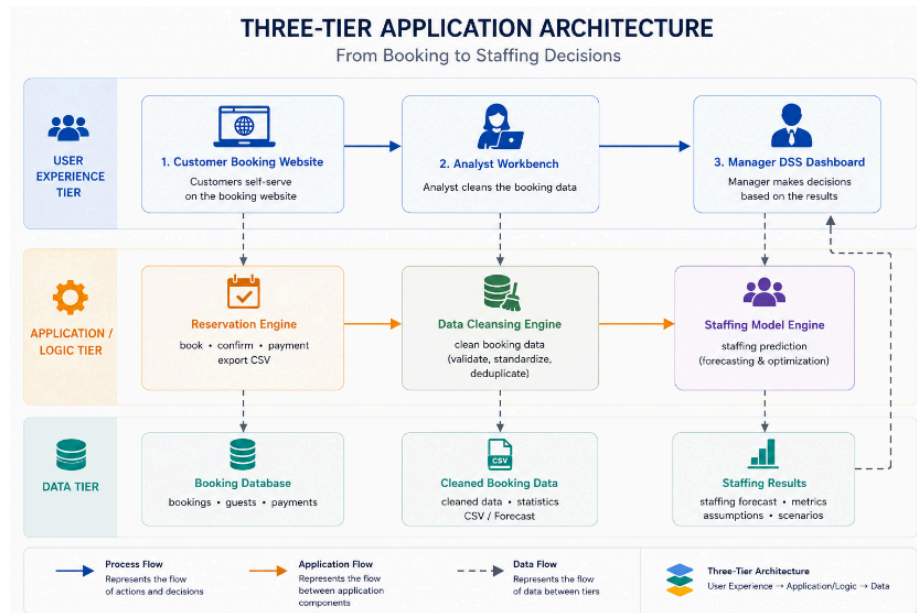


Figure 4: Three-tier architecture: user, logic, and data.

5.1 How the System Works

- **System Input & Output:** The system takes booking and assumption input, processes them through the reservation and decision engines, and returns staffing, cost, and recommendation outputs.



Detailed system tiers:

- **User Stage:** The customer accesses the booking website, selects the cruise itinerary (day trip, half day, 7-night, or 9-night cruise), chooses optional services such as meal plans and transportation, registers an account, selects a payment method, and completes payment. Once the reservation is confirmed, the booking record is stored in the reservation database and included in the demand dataset used for forecasting.
- **Analyst Stage:** Booking records are extracted from the reservation database from different admin tabs and prepared for analysis. The analyst cleans and validates the data, removes duplicates, and performs demand forecasting by cruise type and week, ship utilization, reservation status, total payments, and seasonality analysis. The resulting forecast is then passed on to the staffing model to estimate required staffing levels and expected operating costs.
- **Staffing Stage:** For each cruise type, the forecasted booking volume is multiplied by its average handling time (8, 22, and 28 minutes for day, 7-night, and 9-night cruises, respectively) to calculate the total workload. Workloads across all cruise types are aggregated and converted into weekly labor hours. Required staffing is then calculated based on available working hours per agent, as per these formulas:

$$\text{Weekly Workload Hours} = \frac{\sum(\text{Forecast Volume}_i \times \text{Handle Time}_i)}{60} \quad \text{Required Agents} = \frac{\text{Weekly Workload Hours}}{\text{Available Hours per Agent}}$$

- **Cost Estimation:** The staffing model multiplies the required labor hours by the fully loaded hourly labor rate to estimate weekly and annual operating costs, and predict the number of agents needed to cover the demand.

Manager stage: The manager reviews demand forecasts, staffing requirements, and operating costs through the DSS summary dashboard. The system compares

predicted staffing requirements against the available in-house capacity of 8–12 agents and generates recommendations. If forecast demand can be handled within available capacity, bookings are serviced entirely in-house. If demand exceeds available capacity, the DSS estimates the overflow volume and recommends outsourcing only the excess bookings to third-party agents or adjusting prices to influence demand. During periods of high demand, **DSS may recommend price** increases to maximize profitability, while during low-demand periods it may suggest discounts or promotional offers to improve occupancy.

6. Case Study

To show the prototype in action, we run two weeks of sample data through it: a peak-demand week that exceeds in-house capacity, and a low-demand week that does not.

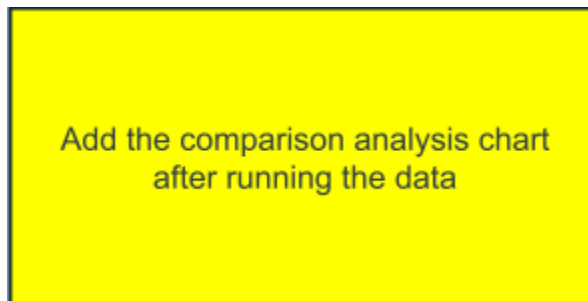


Figure 6. Case study: peak week triggers overflow/pricing; low week triggers discounts.

6.1 Week A_Peak demand

The booking website records **630 bookings** for the week (300 day-cruises, 200 seven-night, 130 nine-night). The analyst exports the CSV; cleaning and forecasting confirm ~174 workload hours. The staffing model predicts **14 agents** are needed, two above the 12-agent cap. The DSS raises an overflow alert. The manager chooses to route roughly two agents' worth of bookings to third-party agents for that week (paying

commission only on that excess) or to raise prices on the highest-demand sailings to moderate volume. Either way, the decision is made on data, not guesswork.

6.2 Week B_ Low demand

A later week records only **280 bookings** (~73 workload hours). The model predicts **7 agents** within the 8–12 in-house range, with spare capacity. No third-party overflow is needed, so zero commission is paid. Seeing slack demand, the DSS recommends discounts or promotional offers to fill low-demand sailings, which the manager can approve to lift bookings and revenue.

7. Key Assumptions

8. Risk Assessment

9. Control Plan

10. Conclusion & Recommendations